Magic Gems: A Polyhedral Framework for Magic Squares

Connecting Combinatorics, Geometry, and Linear Algebra

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1 Magic squares as orthogonality conditions on a height function

Throughout, $n \geq 3$, $N = n^2$, and a cell of the $n \times n$ grid is a pair $c = (i, j)$ with $0 \leq i, j \leq n-1$. An *arrangement* is a bijection $S = (s_{ij})$ from the cells onto $\{1, \ldots, N\}$. We use the centred coordinates of Definition **??**,

$$x_{ij} = j - \tfrac{n-1}{2}, \qquad y_{ij} = \tfrac{n-1}{2} - i, \qquad z_{ij} = s_{ij} - \tfrac{N+1}{2},$$

and refer to $z = z(S)$ as the *height function* of S. A *line* is one of the n rows, the n columns, the main diagonal $\{i = j\}$ or the anti-diagonal $\{i + j = n - 1\}$; there are $2n + 2$ lines, each containing exactly n cells. With $M(n) = n(N+1)/2$ the magic constant, the *deviation* of a line L is

$$\delta_L(S) \;=\; \sum_{c \in L} s_c - M(n) \;=\; \sum_{c \in L} z_c, \tag{1}$$

the second equality because $|L| = n$ and $M(n) = n \cdot \frac{N+1}{2}$. S is *semi-magic* if $\delta_L = 0$ for the $2n$ rows and columns, and *magic* if $\delta_L = 0$ for all $2n + 2$ lines. We write $\rho_k = \delta_{\text{row } k}$, $\gamma_\ell = \delta_{\text{col } \ell}$, δ_{m} and δ_{a} for the row, column, main-diagonal and anti-diagonal deviations.

1.1 Height functions, the inner product, and the energies

Let $\mathcal{F} = \mathbb{R}^{\text{cells}}$ be the N-dimensional space of real functions on the grid, with the normalised inner product and norm

$$\langle f, g \rangle = \frac{1}{N} \sum_{i,j} f_{ij} g_{ij}, \qquad \|f\|^2 = \langle f, f \rangle. \tag{2}$$

The mean of f is $\bar{f} = \langle f, \mathbf{1} \rangle$ and its centring is $\tilde{f} = f - \bar{f}\mathbf{1}$. For two functions ("test function" φ and height f) the empirical covariance over the N cells is $\mathrm{Cov}(\varphi, f) = \langle \tilde{\varphi}, \tilde{f} \rangle = \langle \tilde{\varphi}, f \rangle = \langle \varphi, \tilde{f} \rangle$; the last two equalities hold because $\tilde{\varphi} \perp \mathbf{1}$ and $\tilde{f} \perp \mathbf{1}$.

Lemma 1.1 (Means, variances, and covariance as an inner product)**.** *For every arrangement S of $\{1, \ldots, N\}$:*

1. *$\bar{x} = \bar{y} = \bar{z} = 0$, so z is centred and $\mathrm{Cov}(\varphi, Z) = \langle \tilde{\varphi}, z \rangle = \langle \varphi, z \rangle$ for every $\varphi \in \mathcal{F}$;*
2. $\mathrm{Var}(X) = \mathrm{Var}(Y) = \dfrac{n^2 - 1}{12}, \quad \mathrm{Var}(Z) = \dfrac{N^2 - 1}{12} = \dfrac{n^4 - 1}{12}, \quad \mathrm{Cov}(X, Y) = 0;$
3. *for every line L with indicator $\mathbf{1}_L$,* $\mathrm{Cov}(\mathbf{1}_L, Z) = \dfrac{\delta_L(S)}{N}.$

Proof. (1) $\sum_{i,j} x_{ij} = n\sum_{j=0}^{n-1}\big(j-\frac{n-1}{2}\big) = 0$, and likewise for y; $\sum_{i,j} z_{ij} = \sum_{v=1}^{N} v - N\frac{N+1}{2} = 0$. Since $\bar z = 0$, $\mathrm{Cov}(\varphi, Z) = \langle\tilde\varphi, z\rangle = \langle\varphi, z\rangle - \bar\varphi\langle\mathbf{1}, z\rangle = \langle\varphi, z\rangle$.

(2) $\mathrm{Var}(X) = \frac{1}{N}\sum_{i,j} x_{ij}^2 = \frac{1}{n}\sum_{j=0}^{n-1}\big(j-\frac{n-1}{2}\big)^2$, which is the variance of the uniform distribution on n equally spaced points, $(n^2-1)/12$; the same for Y. $\mathrm{Var}(Z) = \frac{1}{N}\sum_{v=1}^{N}\big(v-\frac{N+1}{2}\big)^2 = (N^2-1)/12$ by the same formula with N points. Finally $\sum_{i,j} x_{ij}y_{ij} = \big(\sum_j x_{\cdot j}\big)\big(\sum_i y_{i\cdot}\big) = 0$ because x depends only on j and y only on i.

(3) By (1), $\mathrm{Cov}(\mathbf{1}_L, Z) = \langle\mathbf{1}_L, z\rangle = \frac{1}{N}\sum_{c\in L} z_c = \delta_L/N$ by (1). □

We write $R_k = \mathbf{1}_{\{i=k\}}$, $C_\ell = \mathbf{1}_{\{j=\ell\}}$, $D_{\mathrm{m}} = \mathbf{1}_{\{i=j\}}$, $D_{\mathrm{a}} = \mathbf{1}_{\{i+j=n-1\}}$ for the $2n+2$ line indicators.

Definition 1.2 (Complete and low-mode energies)**.** For an arrangement S with height function z,

$$E_{\mathrm{full}}(S) = \sum_{k=0}^{n-1}\mathrm{Cov}(R_k, Z)^2 + \sum_{\ell=0}^{n-1}\mathrm{Cov}(C_\ell, Z)^2 + \mathrm{Cov}(D_{\mathrm{m}}, Z)^2 + \mathrm{Cov}(D_{\mathrm{a}}, Z)^2, \tag{3}$$

$$E_{\mathrm{low}}(S) = \mathrm{Cov}(X, Z)^2 + \mathrm{Cov}(Y, Z)^2 + \mathrm{Cov}(D_{\mathrm{m}}, Z)^2 + \mathrm{Cov}(D_{\mathrm{a}}, Z)^2. \tag{4}$$

Note that (3) sums over *all* n rows and *all* n columns. By Lemma 1.1(3) it has the following elementary description, which we use constantly:

$$N^2 E_{\mathrm{full}}(S) = \sum_L \delta_L(S)^2 = \sum_k \rho_k^2 + \sum_\ell \gamma_\ell^2 + \delta_{\mathrm{m}}^2 + \delta_{\mathrm{a}}^2, \tag{5}$$

the sum of the squared line-sum deviations over all $2n+2$ lines. In particular $N^2E_{\mathrm{full}}(S)$ is a non-negative integer.

The dihedral action. The dihedral group D_4 (order 8) acts on the grid by the rotations and reflections of the square; each $g \in D_4$ is a bijection of the set of cells. It acts on arrangements by $(gS)_c = s_{g^{-1}c}$ and on functions by $(gf)_c = f_{g^{-1}c}$, so that $z(gS) = g\,z(S)$ and $\langle gf, gh\rangle = \langle f, h\rangle$. Every g maps lines to lines: it permutes the set of $2n$ rows and columns among themselves and permutes the two diagonals among themselves. Moreover, g acts on the centred coordinates by a signed permutation: $gx \in \{\pm x, \pm y\}$ and $gy \in \{\pm x, \pm y\}$ with $\{gx, gy\} = \{\pm x, \pm y\}$ as unordered pairs up to sign (a rotation by 90° sends $(x, y) \mapsto (-y, x)$, a reflection sends $(x, y) \mapsto (-x, y)$, and so on).

Proposition 1.3 (D_4-invariance of the energies)**.** *For every $g \in D_4$ and every arrangement S: $E_{\mathrm{full}}(gS) = E_{\mathrm{full}}(S)$ and $E_{\mathrm{low}}(gS) = E_{\mathrm{low}}(S)$.*

Proof. For any line L, $\delta_{gL}(gS) = \sum_{c\in gL}(gS)_c = \sum_{c\in L} s_c = \delta_L(S)$ after the substitution $c \mapsto gc$. Since $L \mapsto gL$ is a bijection of the set of all $2n+2$ lines, the multiset $\{\delta_L(gS)\}_L$ equals $\{\delta_L(S)\}_L$ and (5) gives $E_{\mathrm{full}}(gS) = E_{\mathrm{full}}(S)$. For E_{low}: $\mathrm{Cov}(X, gZ) = \langle x, gz\rangle = \langle g^{-1}x, z\rangle$ and $g^{-1}x \in \{\pm x, \pm y\}$, and similarly for y, with $\{g^{-1}x, g^{-1}y\}$ being $\{\pm x, \pm y\}$ in some order; hence $\mathrm{Cov}(X, gZ)^2 + \mathrm{Cov}(Y, gZ)^2 = \mathrm{Cov}(X, Z)^2 + \mathrm{Cov}(Y, Z)^2$. Since g permutes the two diagonals, $\mathrm{Cov}(D_{\mathrm{m}}, gZ)^2 + \mathrm{Cov}(D_{\mathrm{a}}, gZ)^2$ is invariant as well. □

Remark 1.4 (Why the $v4$ definition was not D_4-invariant). The previous version of this paper omitted the last row and the last column from (3), i.e. used $E_{v4} = E_{\mathrm{full}} - \big(\rho_{n-1}^2 + \gamma_{n-1}^2\big)/N^2$, on the grounds that $\sum_k \rho_k = \sum_\ell \gamma_\ell = 0$ makes one row and one column "redundant". Redundant for the *zero set*, yes; but not for the *values*: the horizontal reflection $h : (i, j) \mapsto (i, n-1-j)$ maps column $n-1$ to column 0 and fixes the set of rows, so $E_{v4}(hS) - E_{v4}(S) = \big(\gamma_{n-1}^2 - \gamma_0^2\big)/N^2$, which is nonzero whenever $|\gamma_0| \neq |\gamma_{n-1}|$. Example: swapping the entries $1, 2$ in the first row of the 4×4 magic square $\begin{smallmatrix}1&2&15&16\\12&14&3&5\\13&7&10&4\\8&11&6&9\end{smallmatrix}$ gives $\gamma = (1, -1, 0, 0)$ and $\delta_{\mathrm{m}} = 1$, so $N^2E_{\mathrm{full}} = 3$ for the whole D_4 orbit while N^2E_{v4} takes the values 2 and 3 on it. All E_{full} statistics in this paper use the symmetric definition (3).

1.2 The projection lemma

Let $\tilde{R}_k = R_k - \frac{1}{n}\mathbf{1}$, $\tilde{C}_\ell = C_\ell - \frac{1}{n}\mathbf{1}$, $\tilde{D}_{\mathrm{m}} = D_{\mathrm{m}} - \frac{1}{n}\mathbf{1}$, $\tilde{D}_{\mathrm{a}} = D_{\mathrm{a}} - \frac{1}{n}\mathbf{1}$ be the centred line indicators (each line has n of the N cells, so each mean is $1/n$), and set

$$W_R = \operatorname{span}\{\tilde{R}_k\}, \quad W_C = \operatorname{span}\{\tilde{C}_\ell\}, \quad W_D = \operatorname{span}\{\tilde{D}_{\mathrm{m}}, \tilde{D}_{\mathrm{a}}\}, \quad W = W_R + W_C + W_D \subset \mathbf{1}^\perp.$$

Lemma 1.5 (Structure of W). *For $n \geq 3$:*

1. *$W_R = \{f : f_{ij} = a_i,\ \sum_i a_i = 0\}$ and $W_C = \{f : f_{ij} = b_j,\ \sum_j b_j = 0\}$; both have dimension $n-1$.*
2. *$\langle \tilde{R}_k, \tilde{R}_{k'} \rangle = \frac{1}{N}(n\delta_{kk'} - 1)$, and the same for columns.*
3. *$W_R \perp W_C$, $W_R \perp W_D$, $W_C \perp W_D$.*
4. *$\langle \tilde{D}_{\mathrm{m}}, \tilde{D}_{\mathrm{m}} \rangle = \langle \tilde{D}_{\mathrm{a}}, \tilde{D}_{\mathrm{a}} \rangle = \frac{n-1}{N}$ and $\langle \tilde{D}_{\mathrm{m}}, \tilde{D}_{\mathrm{a}} \rangle = \frac{\epsilon_n}{N}$ with $\epsilon_n = 0$ for n odd and $\epsilon_n = -1$ for n even; in particular $\tilde{D}_{\mathrm{m}}, \tilde{D}_{\mathrm{a}}$ are linearly independent and $\dim W_D = 2$.*
5. *$W = W_R \oplus W_C \oplus W_D$ is an orthogonal direct sum and $\dim W = 2n$.*

Proof. For two lines L, L', $N\langle \tilde{\mathbf{1}}_L, \tilde{\mathbf{1}}_{L'} \rangle = \sum_c (\mathbf{1}_L - \frac{1}{n})(\mathbf{1}_{L'} - \frac{1}{n}) = |L \cap L'| - \frac{n}{n} - \frac{n}{n} + \frac{N}{n^2} = |L \cap L'| - 1$. Two distinct rows are disjoint and a row meets itself in n cells, giving (2). A row meets every column and each diagonal in exactly one cell, and so does a column; hence all those inner products vanish, which is (3). A diagonal meets itself in n cells; the two diagonals meet in one cell (the centre) when n is odd and in none when n is even, giving (4); the Gram matrix $\frac{1}{N}\left(\begin{smallmatrix} n-1 & \epsilon_n \\ \epsilon_n & n-1 \end{smallmatrix}\right)$ has determinant $\big((n-1)^2 - \epsilon_n^2\big)/N^2 > 0$ for $n \geq 3$, so the two vectors are independent. For (1): $\sum_k a_k \tilde{R}_k$ is the function $(i,j) \mapsto a_i - \bar{a}$, a function of i alone with zero mean, and every such function arises (take a itself); the map $a \mapsto a - \bar{a}$ has rank $n-1$. Finally (5) follows from (1), (3) and (4): the three subspaces are mutually orthogonal, hence independent, and $\dim W = (n-1) + (n-1) + 2 = 2n$. □

Remark 1.6. E_{full} is a sum of $2n+2$ squares but $\dim W = 2n$: the two identities $\sum_k \operatorname{Cov}(R_k, Z) = \sum_\ell \operatorname{Cov}(C_\ell, Z) = 0$ (equivalently $\sum_k \tilde{R}_k = \sum_\ell \tilde{C}_\ell = 0$) are the only linear relations among the centred indicators. Dropping one row and one column removes the redundancy at the price of D_4-invariance (Remark 1.4); the symmetric sum keeps both.

For a subspace $V \subset \mathcal{F}$ let P_V denote the orthogonal projection with respect to (2).

Lemma 1.7 (Projection lemma). *Let S be an arrangement with height function z, and let $\rho, \gamma \in \mathbb{R}^n$ and $\delta = (\delta_{\mathrm{m}}, \delta_{\mathrm{a}}) \in \mathbb{R}^2$ be its line deviations. Then*

1. *$P_{W_R} z$ is the function $(i,j) \mapsto \rho_i/n$ (the row means of z), $P_{W_C} z$ is $(i,j) \mapsto \gamma_j/n$, and*

$$\sum_k \operatorname{Cov}(R_k, Z)^2 = \frac{\|\rho\|^2_{\mathbb{R}^n}}{N^2} = \frac{1}{n}\|P_{W_R} z\|^2, \qquad \sum_\ell \operatorname{Cov}(C_\ell, Z)^2 = \frac{\|\gamma\|^2_{\mathbb{R}^n}}{N^2} = \frac{1}{n}\|P_{W_C} z\|^2.$$

2. *With $G = \left(\begin{smallmatrix} n-1 & \epsilon_n \\ \epsilon_n & n-1 \end{smallmatrix}\right)$ as in Lemma 1.5(4), $\|P_{W_D} z\|^2 = \frac{1}{N}\,\delta^{\mathsf{T}} G^{-1} \delta$, while $\operatorname{Cov}(D_{\mathrm{m}}, Z)^2 + \operatorname{Cov}(D_{\mathrm{a}}, Z)^2 = \delta^{\mathsf{T}}\delta/N^2$.*
3. *Consequently*

$$E_{\mathrm{full}}(S) = \frac{1}{n}\Big(\|P_{W_R} z\|^2 + \|P_{W_C} z\|^2\Big) + \frac{\delta^{\mathsf{T}}\delta}{N^2} = Q\big(P_W z\big), \tag{6}$$

where Q is the positive-definite quadratic form on W given by $Q(w) = \frac{1}{n}\|P_{W_R} w\|^2 + \frac{1}{n}\|P_{W_C} w\|^2 + \frac{1}{N^2}\, w_D^{\mathsf{T}} G^2\, w_D$, $w_D \in \mathbb{R}^2$ denoting the coordinates of $P_{W_D} w$ in the basis $(\tilde{D}_{\mathrm{m}}, \tilde{D}_{\mathrm{a}})$ (so that $\|P_{W_D} w\|^2 = \frac{1}{N} w_D^{\mathsf{T}} G w_D$). In particular E_{full} depends on z only through $P_W z$, and

$$\frac{n - 1 - |\epsilon_n|}{n^2}\, \|P_W z\|^2 \;\leq\; E_{\mathrm{full}}(S) \;\leq\; \frac{1}{n}\, \|P_W z\|^2. \tag{7}$$

4. *$E_{\text{full}} \neq \|P_W z\|^2$ in general: the weights $1/n$ on the row and column components and $(n-1\pm|\epsilon_n|)/N$ on the diagonal components differ, because the centred indicators are not an orthonormal system (Lemma 1.5(2),(4)). For odd n the diagonal part is exactly $\frac{n-1}{N}\|P_{W_D} z\|^2$.*

Proof. (1) By Lemma 1.5(1), W_R is the space of zero-mean functions of i alone. The orthogonal projection of z onto the space of all functions of i alone is the function of row means, $(i,j) \mapsto \frac{1}{n}\sum_j z_{ij} = \rho_i/n$ by (1); this already has zero mean because z does (Lemma 1.1(1)), so it equals $P_{W_R} z$. Its squared norm is $\frac{1}{N}\sum_i n(\rho_i/n)^2 = \|\rho\|^2/n^3$, while $\sum_k \mathrm{Cov}(R_k, Z)^2 = \sum_k \rho_k^2/N^2 = \|\rho\|^2/n^4$ by Lemma 1.1(3). Columns are identical with $i \leftrightarrow j$.

(2) Write $\Phi = (\tilde{D}_{\mathrm{m}}, \tilde{D}_{\mathrm{a}})$. The Gram matrix of Φ is G/N and $\langle \tilde{D}_{\mathrm{m}}, z\rangle = \mathrm{Cov}(D_{\mathrm{m}}, Z) = \delta_{\mathrm{m}}/N$, so the vector of inner products is δ/N. The standard formula $\|P_{\mathrm{span}\,\Phi} z\|^2 = b^{\mathsf{T}}\Gamma^{-1} b$ with Gram matrix $\Gamma = G/N$ and $b = \delta/N$ gives $\|P_{W_D} z\|^2 = \frac{1}{N}\delta^{\mathsf{T}} G^{-1}\delta$. The second statement is Lemma 1.1(3).

(3) Summing (1) and (2) gives the first equality in (6). Since the three subspaces are orthogonal (Lemma 1.5(5)), $P_{W_R} z = P_{W_R} P_W z$ etc. If $P_{W_D} z = w_{\mathrm{m}}\tilde{D}_{\mathrm{m}} + w_{\mathrm{a}}\tilde{D}_{\mathrm{a}}$ then $\delta_{\mathrm{m}}/N = \langle \tilde{D}_{\mathrm{m}}, z\rangle = \langle \tilde{D}_{\mathrm{m}}, P_{W_D} z\rangle = \big((n-1)w_{\mathrm{m}} + \epsilon_n w_{\mathrm{a}}\big)/N$ and similarly for δ_{a}, i.e. $\delta = G\, w_D$; thus $\delta^{\mathsf{T}}\delta/N^2 = w_D^{\mathsf{T}} G^2 w_D/N^2$, which is the third piece of Q. Each piece of Q is positive definite on its orthogonal summand (G^2 is positive definite because G is invertible), so Q is positive definite on W. For the bounds, $\delta^{\mathsf{T}}\delta$ lies between $\lambda_{\min}(G)\,\delta^{\mathsf{T}} G^{-1}\delta$ and $\lambda_{\max}(G)\,\delta^{\mathsf{T}} G^{-1}\delta$, and the eigenvalues of G are $n-1\pm\epsilon_n$, i.e. $n-1\mp|\epsilon_n|$; hence the diagonal part of E_{full} lies between $\frac{n-1-|\epsilon_n|}{N}\|P_{W_D} z\|^2$ and $\frac{n-1+|\epsilon_n|}{N}\|P_{W_D} z\|^2$. Since $\frac{n-1+|\epsilon_n|}{n^2} \le \frac{n}{n^2} = \frac{1}{n}$ and $\|P_W z\|^2 = \|P_{W_R} z\|^2 + \|P_{W_C} z\|^2 + \|P_{W_D} z\|^2$ by orthogonality, (7) follows. (4) is a restatement of the weights computed above. □

Lemma 1.8 (Zero set of the energies)**.** *For every arrangement S of $\{1,\dots,N\}$:*

1. *$E_{\text{full}}(S) = 0 \iff P_W z = 0 \iff z \perp W \iff S$ is a magic square.*
2. *$\sum_k \mathrm{Cov}(R_k, Z)^2 + \sum_\ell \mathrm{Cov}(C_\ell, Z)^2 = 0 \iff z \perp W_R \oplus W_C \iff S$ is semi-magic.*
3. *$E_{\text{low}}(S) = 0 \implies$ both diagonals sum to $M(n)$; the implication "$E_{\text{low}}(S) = 0 \implies S$ magic" holds for $n = 3$ and fails for $n = 4$ (Remark 1.16).*

Proof. (1) $E_{\text{full}}(S) = 0$ iff $Q(P_W z) = 0$ iff $P_W z = 0$ (positive definiteness) iff $z \perp W$. By (5), $E_{\text{full}}(S) = 0$ iff all $2n+2$ line deviations vanish, i.e. iff S is magic. (2) is the same argument with the diagonal terms removed. (3) The two diagonal terms of E_{low} are $\delta_{\mathrm{m}}^2/N^2$ and $\delta_{\mathrm{a}}^2/N^2$; the rest is Remark 1.16. □

Remark 1.9 (What the projection lemma does and does not say). Lemma 1.8(1) is the definition of a magic square rewritten in the language of orthogonality: "all line sums equal $M(n)$" is literally "$z \perp \tilde{\mathbf{1}}_L$ for every line L". It is not a theorem about magic squares, and we do not present it as one. Its use is as a change of viewpoint. Once the constraints are read as orthogonality of the height function to a $2n$-dimensional test space W, three things become available that are not visible in the line-sum formulation: (i) W can be truncated or enlarged in natural ways (polynomial test functions, Section 1.3), which produces a hierarchy of relaxations with Loly's inertia invariance as the degree-two member; (ii) for a uniformly random arrangement the quantities $\langle \varphi, z\rangle$ are permutation statistics in the sense of Hoeffding, whose exact moments and limiting law are computable (see the section on random arrangements);

(iii) a single swap changes z by a multiple of a difference of two cell indicators, and (5) turns the change of E_{full} into a two-line count (Section 1.4).

1.3 The moment hierarchy

The functions x and y are the degree-one polynomials in the column and row index. Replacing them by higher powers gives a natural family of relaxations of the semi-magic condition.

Definition 1.10 (Degree-k conditions)**.** Let $f \in \mathcal{F}$ be centred ($\langle f, \mathbf{1}\rangle = 0$) and $k \geq 1$. We say f satisfies the *degree-k conditions* if

$$\langle x^m, f\rangle = 0 \quad \text{and} \quad \langle y^m, f\rangle = 0 \qquad \text{for all } 1 \leq m \leq k.$$

For $f = z(S)$ these read $\mathrm{Cov}(X^m, Z) = \mathrm{Cov}(Y^m, Z) = 0$ for $1 \leq m \leq k$, by Lemma 1.1(1).

Write $x_j = j - \frac{n-1}{2}$ and $y_i = \frac{n-1}{2} - i$ for the n distinct values taken by x and y, and for centred f let $\gamma_j(f) = \sum_i f_{ij}$ and $\rho_i(f) = \sum_j f_{ij}$ be its column and row sums (for $f = z(S)$ these are the deviations γ, ρ of (1)).

Theorem 1.11 (Moment hierarchy)**.** *Let $n \geq 3$ and let $f \in \mathcal{F}$ be centred.*

(a) For every function φ that depends only on the column index, $\langle \varphi, f\rangle = \frac{1}{N}\sum_j \varphi(j)\gamma_j(f)$; in particular $\langle x^m, f\rangle = \frac{1}{N}\sum_j x_j^m \gamma_j(f)$ and $\langle y^m, f\rangle = \frac{1}{N}\sum_i y_i^m \rho_i(f)$ for all $m \geq 0$. Hence if $\gamma(f) = 0$ and $\rho(f) = 0$ (for $f = z(S)$: if S is semi-magic), then $\langle x^m, f\rangle = \langle y^m, f\rangle = 0$ for every *$m \geq 0$.*

(b) If $k \geq n-1$, the degree-k conditions imply $\gamma(f) = \rho(f) = 0$.

(c) If $1 \leq k \leq n-2$, there is a centred f satisfying the degree-k conditions with $\gamma(f) \neq 0$; explicitly, $f_{ij} = \frac{1}{n}(-1)^j\binom{k+1}{j}$ for $0 \leq j \leq k+1$ and $f_{ij} = 0$ for $j > k+1$. This f violates the degree-$(k+1)$ condition.

Thus, for real-valued height functions, the degree-k conditions are equivalent to "equal row sums and equal column sums" exactly when $k \geq n-1$.

Proof. (a) $\langle \varphi, f\rangle = \frac{1}{N}\sum_j \varphi(j)\sum_i f_{ij} = \frac{1}{N}\sum_j \varphi(j)\gamma_j(f)$; the row statement is symmetric. The last sentence is immediate.

(b) Suppose $\langle x^m, f\rangle = 0$ for $1 \leq m \leq k$ with $k \geq n-1$. Together with $\sum_j \gamma_j(f) = \sum_{i,j} f_{ij} = 0$ (the case $m = 0$), part (a) says that $\gamma(f) \in \mathbb{R}^n$ is orthogonal to the n vectors $(x_j^m)_j$, $m = 0, \ldots, n-1$. These are the columns of the Vandermonde matrix of the n *distinct* nodes $x_0, \ldots, x_{n-1}$, which is invertible; hence they span $\mathbb{R}^n$ and $\gamma(f) = 0$. The same argument with the nodes y_i gives $\rho(f) = 0$.

(c) Let $v_j = (-1)^j\binom{k+1}{j}$ for $0 \leq j \leq k+1$ and $v_j = 0$ for $k+1 < j \leq n-1$ (this needs $k+1 \leq n-1$). For a polynomial p define the forward difference $(\Delta p)(t) = p(t+1) - p(t)$; Δ lowers the degree of a nonconstant polynomial by exactly one and kills constants, so $\Delta^{k+1}p = 0$ whenever $\deg p \leq k$. Writing E for the shift $(Ep)(t) = p(t+1)$ we have $\Delta = E - 1$, and since E and 1 commute the binomial theorem gives $\Delta^{k+1} = \sum_{j=0}^{k+1}\binom{k+1}{j}(-1)^{k+1-j}E^j$, i.e.

$$(\Delta^{k+1}p)(0) = (-1)^{k+1}\sum_{j=0}^{k+1}(-1)^j\binom{k+1}{j}p(j) = (-1)^{k+1}\sum_{j=0}^{n-1} v_j\, p(j). \tag{8}$$

Taking $p(t) = (t - \frac{n-1}{2})^m$ with $0 \leq m \leq k$ shows $\sum_j v_j x_j^m = 0$; in particular ($m = 0$) $\sum_j v_j = 0$. Now set $f_{ij} = v_j/n$. Then $\sum_{i,j} f_{ij} = \sum_j v_j = 0$, so f is centred; $\rho_i(f) = \frac{1}{n}\sum_j v_j = 0$ for every i, so all $\langle y^m, f\rangle$ vanish by (a); $\gamma_j(f) = v_j$, so $\langle x^m, f\rangle = \frac{1}{N}\sum_j v_j x_j^m = 0$ for $1 \leq m \leq k$ by (8); and $\gamma_0(f) = v_0 = 1 \neq 0$. Finally, for $p(t) = (t - \frac{n-1}{2})^{k+1}$ we have $\Delta^{k+1}p \equiv (k+1)! \neq 0$, so $\langle x^{k+1}, f\rangle = \pm(k+1)!/N \neq 0$. □

Corollary 1.12 (E_{low} is the degree-one truncation)**.** *$E_{\mathrm{low}}(S) = \langle x, z\rangle^2 + \langle y, z\rangle^2 + \big(\delta_{\mathrm{m}}^2 + \delta_{\mathrm{a}}^2\big)/N^2$: its first two terms are the degree-1 conditions of Definition 1.10 in squared form, and $E_{\mathrm{low}}(S) = 0$ iff z satisfies the degree-1 conditions and both diagonals sum to $M(n)$. By Theorem 1.11(a), every magic square has $E_{\mathrm{low}} = 0$; by Theorem 1.11(c) the degree-1 conditions characterise equal line sums (among real height functions) only when $n - 1 \le 1$, i.e. never for $n \ge 3$.*

Proof. Immediate from Lemma 1.1(1),(3) and Theorem 1.11. □

Corollary 1.13 (Loly's moment-of-inertia invariance is the degree-two statement)**.** *Place a mass s_{ij} at the planar point (x_{ij}, y_{ij}) and let $I(S) = \sum_{i,j} s_{ij}\big(x_{ij}^2 + y_{ij}^2\big)$ be the moment of inertia about the centre. Then for every arrangement*

$$I(S) = N\,\mathrm{Cov}\big(Z,\, X^2 + Y^2\big) + \frac{N+1}{2}\sum_{i,j}\big(x_{ij}^2 + y_{ij}^2\big) = \sum_j x_j^2\gamma_j + \sum_i y_i^2\rho_i + \frac{n^2(n^2+1)(n^2-1)}{12}. \tag{9}$$

Consequently $I(S) = \dfrac{n^2(n^4-1)}{12}$ for every semi-magic square S (no diagonal condition is needed), and more generally for every arrangement satisfying the degree-2 conditions. The values $60, 340, 1300, 3885, 9800, 21840, 44280$ for $n = 3, \dots, 9$ are OEIS A126275 (Loly [1]).

Proof. Substitute $s_{ij} = z_{ij} + \frac{N+1}{2}$: $I(S) = \sum_{i,j} z_{ij}(x_{ij}^2 + y_{ij}^2) + \frac{N+1}{2}\sum_{i,j}(x_{ij}^2 + y_{ij}^2)$. The first sum is $N\langle x^2 + y^2, z\rangle = N\mathrm{Cov}(X^2 + Y^2, Z)$ by Lemma 1.1(1), and equals $\sum_j x_j^2\gamma_j + \sum_i y_i^2\rho_i$ by Theorem 1.11(a). For the second, $\sum_{i,j} x_{ij}^2 = N\mathrm{Var}(X) = n^2\frac{n^2-1}{12}$ and the same for y (Lemma 1.1(2)), so $\frac{N+1}{2}\cdot 2\cdot\frac{n^2(n^2-1)}{12} = \frac{n^2(n^2+1)(n^2-1)}{12} = \frac{n^2(n^4-1)}{12}$. If S satisfies the degree-2 conditions—in particular if S is semi-magic, by Theorem 1.11(a)—the first two terms vanish. □

Theorem 1.11 is a statement about real-valued height functions. For arrangements, i.e. for $f = z(S)$ with S a permutation of $\{1, \dots, N\}$, the integrality and boundedness of the line sums change the picture in an interesting way.

Proposition 1.14 (The hierarchy for permutations)**.** *For $n \ge 3$ let $k^*(n)$ be the least k such that every arrangement S of $\{1, \dots, N\}$ satisfying the degree-k conditions is semi-magic. Then:*

1. *$k^*(n) \le n - 1$ for all n.*
2. *$k^*(3) = 2$ and $k^*(4) = 3$. Witnesses for the lower bounds:*

$$n = 3,\ \textit{degree } 1: \quad \begin{pmatrix} 1 & 3 & 9 \\ 8 & 6 & 5 \\ 7 & 4 & 2 \end{pmatrix}, \qquad \gamma = (1, -2, 1),\ \rho = (-2, 4, -2);$$

$$n = 4,\ \textit{degree } 2: \quad \begin{pmatrix} 14 & 2 & 5 & 13 \\ 8 & 15 & 7 & 4 \\ 12 & 3 & 9 & 10 \\ 1 & 11 & 16 & 6 \end{pmatrix}, \qquad \gamma = (1, -3, 3, -1),\ \rho = 0.$$

3. *$k^*(n) \le n - 2$ for all $n \ge 14$: for such n the degree-$(n-2)$ conditions already force equal row and column sums.*

Proof. (1) is Theorem 1.11(b).

(2) Lower bounds. For the 3×3 array, the column sums are $16, 13, 16$ and the row sums $13, 19, 13$, so $\gamma = (1, -2, 1)$, $\rho = (-2, 4, -2)$ and $\sum_j x_j\gamma_j = (-1)(1) + 0 + (1)(1) = 0$, $\sum_i y_i\rho_i = (1)(-2) + 0 + (-1)(-2) = 0$: the degree-1 conditions hold but the square is not semi-magic, so $k^*(3) \ge 2$. For the 4×4 array, every row sums to 34 and the column sums are $35, 31, 37, 33$, so $\rho = 0$ and $\gamma = (1, -3, 3, -1)$, which is the vector v of Theorem 1.11(c) with $k = 2$; hence

$\sum_j x_j^m \gamma_j = 0$ for $m = 1, 2$ by (8), the degree-2 conditions hold, and the square is not semi-magic, so $k^*(4) \geq 3$. The upper bounds are (1).

(3) Let S satisfy the degree-$(n-2)$ conditions. By Theorem 1.11(a), γ is orthogonal to the $n-1$ linearly independent vectors $(x_j^m)_j$, $m = 0, \ldots, n-2$, so it lies in their one-dimensional orthogonal complement in $\mathbb{R}^n$. By (8) with $k+1 = n-1$, this complement is spanned by $v_j = (-1)^j \binom{n-1}{j}$, $j = 0, \ldots, n-1$ (which is orthogonal to all polynomials of degree $\leq n-2$ and nonzero). So $\gamma = tv$ with $t = \gamma_0 \in \mathbb{Z}$ because $v_0 = 1$ and γ is an integer vector. On the other hand every column sum of a permutation of $\{1, \ldots, N\}$ lies between the sum of the n smallest and the sum of the n largest values, so

$$|\gamma_j| \leq \sum_{v=N-n+1}^{N} v - M(n) = \frac{n(2N-n+1)}{2} - \frac{n(N+1)}{2} = \frac{n(N-n)}{2} = \frac{n^2(n-1)}{2}.$$

If $t \neq 0$ then $\max_j |\gamma_j| \geq \binom{n-1}{\lfloor (n-1)/2 \rfloor}$. We claim $\binom{n-1}{\lfloor (n-1)/2 \rfloor} > \frac{n^2(n-1)}{2}$ for all $n \geq 14$. For $n = 14$: $\binom{13}{6} = 1716 > 1274 = \frac{14^2 \cdot 13}{2}$. Passing from n to $n+1$ multiplies the left side by $\binom{n}{\lfloor n/2 \rfloor} / \binom{n-1}{\lfloor (n-1)/2 \rfloor}$, which equals 2 for n even and $\frac{2n}{n+1} \geq \frac{28}{15}$ for odd $n \geq 14$, while it multiplies the right side by $\frac{(n+1)^2 n}{n^2(n-1)} = \frac{(n+1)^2}{n(n-1)} \leq \frac{225}{182} < \frac{28}{15}$ for $n \geq 14$; the claim follows by induction. Hence $t = 0$ and $\gamma = 0$; the same argument applied to the rows gives $\rho = 0$. □

Remark 1.15 (Values of $k^*(n)$ for $5 \leq n \leq 13$). For $5 \leq n \leq 13$ a randomised swap search (`verify_theory.py`) found, for each n, an arrangement with all row sums equal to $M(n)$ and column deviations $\gamma_j = (-1)^j \binom{n-1}{j}$; such an arrangement satisfies the degree-$(n-2)$ conditions without being semi-magic, so $k^*(n) = n-1$ for all $3 \leq n \leq 13$. Combined with Proposition 1.14(3), the permutation threshold agrees with the linear-algebra threshold $n-1$ exactly for $n \leq 13$ and drops below it from $n = 14$ on; determining $k^*(n)$ for $n \geq 14$ is open. We note this because it illustrates concretely what the "permutation constraint" adds to the orthogonality reformulation: for real heights the test space of degree $\leq n-2$ is too small, whereas for large n the integer heights cannot realise the one direction it misses.

Remark 1.16 (Why the low-mode energy appeared to work for $n = 3$). For $n = 3$ an exhaustive enumeration of all $9! = 362{,}880$ arrangements (`verify_theory.py`, exact integer arithmetic) gives: 760 arrangements satisfy the degree-1 conditions $\mathrm{Cov}(X, Z) = \mathrm{Cov}(Y, Z) = 0$; exactly 72 (the semi-magic squares) satisfy the degree-2 conditions, in accordance with $k^*(3) = 2$; and exactly 8 satisfy the degree-1 conditions *together with* both diagonal conditions—and these 8 are the magic squares. So for $n = 3$, $E_{\mathrm{low}}(S) = 0 \iff S$ is magic, but for the "wrong" reason: the degree-1 conditions leave the one-parameter families $\gamma = t(1, -2, 1)$, $\rho = u(1, -2, 1)$ open, and it is the two diagonal conditions combined with the requirement that the entries be a permutation of $1, \ldots, 9$ that happen to force $t = u = 0$. Nothing of the sort survives for $n = 4$: the arrangement $\begin{smallmatrix} 9 & 4 & 12 & 3 \\ 16 & 1 & 14 & 7 \\ 10 & 15 & 11 & 8 \\ 2 & 5 & 6 & 13 \end{smallmatrix}$ of Remark **??** has $E_{\mathrm{low}} = 0$ (both diagonals sum to 34, $\gamma = (3, -9, 9, -3)$, $\rho = (-6, 4, 10, -8)$) and is not even semi-magic; it satisfies the degree-2 condition in x but not in y ($\sum_i y_i^2 \rho_i = -28$).

1.4 The single-swap formula

Let $a \neq b$ be two cells and let S^{ab} be the arrangement obtained from S by exchanging the entries in a and b. Define

$$m(a, b) = \#\{\text{lines } L : \ L \text{ contains exactly one of } a, b\},$$
$$u_S(c) = \sum_{L \ni c} \delta_L(S) \qquad (\text{sum over the lines through } c).$$

Theorem 1.17 (Single-swap formula). *For every arrangement S and cells $a \neq b$, with $d = s_b - s_a$,*

$$N^2\big(E_{\text{full}}(S^{ab}) - E_{\text{full}}(S)\big) = d^2\, m(a,b) + 2d\,\big(u_S(a) - u_S(b)\big). \tag{10}$$

In particular, if S is a magic square then, with $d = |s_a - s_b|$,

$$E_{\text{full}}(S^{ab}) = \frac{d^2\, m(a,b)}{N^2}. \tag{11}$$

Proof. The swap changes the height function by $z(S^{ab}) = z(S) + d(\mathbf{1}_a - \mathbf{1}_b)$. For a line L, $\delta_L(S^{ab}) = \delta_L(S) + d\big(\mathbf{1}_L(a) - \mathbf{1}_L(b)\big)$: the deviation increases by d if L contains a but not b, decreases by d if L contains b but not a, and is unchanged if L contains both or neither. By (5),

$$\begin{aligned} N^2\Delta E_{\text{full}} &= \sum_{L\ni a,\, L\not\ni b} \big[(\delta_L + d)^2 - \delta_L^2\big] + \sum_{L\ni b,\, L\not\ni a} \big[(\delta_L - d)^2 - \delta_L^2\big] \\ &= d^2 m(a,b) + 2d\Big(\sum_{L\ni a,\, L\not\ni b} \delta_L - \sum_{L\ni b,\, L\not\ni a} \delta_L\Big). \end{aligned}$$

Adding and subtracting the lines containing both a and b turns the bracket into $u_S(a) - u_S(b)$. If S is magic, all $\delta_L(S) = 0$, so $u_S \equiv 0$ and $E_{\text{full}}(S) = 0$, giving (11). □

Lemma 1.18 (The multiplicity $m(a,b)$). *For distinct cells $a = (i,j)$, $b = (i',j')$:*

$$\begin{aligned} m(a,b) = 2\cdot[i \neq i'] + 2\cdot[j \neq j'] &+ [\textit{exactly one of } a,b \textit{ on the main diagonal}] \\ &+ [\textit{exactly one of } a,b \textit{ on the anti-diagonal}]. \end{aligned}$$

Hence $2 \le m(a,b) \le 6$, and $m(a,b) = 2$ if and only if a and b lie in a common row or a common column and neither of them lies on a diagonal. Such pairs exist for every $n \ge 3$ (for $n = 3$ exactly one pair up to D_4: the two non-central cells of the middle row or column; for $n \ge 4$ any two off-diagonal cells of a row).

Proof. Rows: if $i = i'$ the common row contains both cells and no other row contains either; if $i \neq i'$ the row of a contains a only and the row of b contains b only. Columns likewise. Each diagonal contributes 1 or 0 according to whether it separates a from b. Since $a \neq b$, not both $i = i'$ and $j = j'$, so the row/column contribution is 2 or 4 and $m \ge 2$; the maximum 6 is $4 + 1 + 1$. Now $m = 2$ forces $i = i'$ or $j = j'$ (not both) and no diagonal separating a from b. If $i = i'$ and $j \neq j'$, both cells on the main diagonal would mean $j = i = j'$, impossible; so "not separated by the main diagonal" means neither is on it, and likewise for the anti-diagonal. The column case is symmetric. For existence: a row i contains the diagonal cells (i,i) and $(i, n-1-i)$ (which coincide iff n is odd and $i = \frac{n-1}{2}$), so it has at least $n - 2 \ge 2$ off-diagonal cells when $n \ge 4$, and the middle row of the 3×3 grid has the two off-diagonal cells $(1,0), (1,2)$. □

Corollary 1.19 (Strict local minimality and the exact gap). *Let S be a magic square of order $n \ge 3$.*

1. *Every single swap strictly increases the energy: $E_{\text{full}}(S^{ab}) \ge 2/N^2 > 0$ for all $a \neq b$. Thus every magic square is a strict local minimum of E_{full} on the set of arrangements with respect to transpositions, for every n.*

2. *The smallest energy reachable by one swap is*

$$\Delta_{\min}(S) = \min_{a\neq b} \frac{(s_a - s_b)^2\, m(a,b)}{N^2}, \qquad \frac{2}{N^2} \le \Delta_{\min}(S) \le \frac{6}{N^2}.$$

   The lower bound is attained iff some two cells with consecutive values share a row or column and are both off the diagonals; the upper bound follows by swapping any two consecutive values ($d = 1$, $m \le 6$). In particular $\Delta_{\min}(S) = \Theta(n^{-4})$ with explicit constants, uniformly over all magic squares of order n.

3. *The possible values of* $N^2E_{\mathrm{full}}(S^{ab})$ *are the integers* d^2m *with* $1 \le d \le N-1$ *and* $2 \le m \le 6$. *For the* 880 *magic squares of order* 4, *the multiset of* $N^2\Delta_{\min}$ *is* $\{2:432,\ 3:420,\ 4:28\}$, *and for the* 8 *squares of order* 3 *it is* $\{3:8\}$ *(computation; every one of the* 105,888 *single swaps satisfies* (11) *exactly).*

Proof. (1) and (2): by (11), $N^2E_{\mathrm{full}}(S^{ab}) = d^2m(a,b)$ with $d \ge 1$ and $m \ge 2$ (Lemma 1.18). The pair of cells carrying the values $v, v+1$ has $d = 1$ and $m \le 6$, giving the upper bound; the characterisation of equality in the lower bound is Lemma 1.18. (3) is immediate from (11) and the computation reported in Section **??**. □

Remark 1.20. Version 4 of this paper reported a "trimodal" distribution of swap gaps and a $1/n^2$ scaling of the minimum gap. Both were artifacts: the gaps are the integers d^2m divided by $N^2 = n^4$, and the three "modes" were the values $d^2m \in \{2,3,4\}$ (with the asymmetric E_{v4}, $\{1,2,3\}$).

Proposition 1.21 (Swap formula for E_{low})**.** *For every arrangement* S *and cells* $a \ne b$ *with* $d = s_b - s_a$,

$$\mathrm{Cov}(X, Z^{ab}) = \mathrm{Cov}(X,Z) + \frac{d\,(x_a - x_b)}{N}, \qquad \mathrm{Cov}(Y, Z^{ab}) = \mathrm{Cov}(Y,Z) + \frac{d\,(y_a - y_b)}{N},$$
$$\mathrm{Cov}(D, Z^{ab}) = \mathrm{Cov}(D,Z) + \frac{d\,(D(a) - D(b))}{N}$$

for $D \in \{D_{\mathrm{m}}, D_{\mathrm{a}}\}$. *If* $E_{\mathrm{low}}(S) = 0$ *(in particular if* S *is magic),*

$$E_{\mathrm{low}}(S^{ab}) = \frac{d^2}{N^2}\Big[(x_a - x_b)^2 + (y_a - y_b)^2 + \big(D_{\mathrm{m}}(a) - D_{\mathrm{m}}(b)\big)^2 + \big(D_{\mathrm{a}}(a) - D_{\mathrm{a}}(b)\big)^2\Big],$$

i.e. d^2/N^2 *times the squared planar distance between the two cells plus the number of diagonals separating them. Since* $a \ne b$ *the bracket is at least* 1, *so magic squares are strict local minima of* E_{low} *as well, with* $E_{\mathrm{low}}(S^{ab}) \ge 1/N^2$; *the squared planar distance is bounded by* $2(n-1)^2$, *so* $E_{\mathrm{low}}(S^{ab}) \le d^2\big(2(n-1)^2 + 2\big)/N^2$.

Proof. $\mathrm{Cov}(\varphi, Z^{ab}) = \langle \varphi, z + d(\mathbf{1}_a - \mathbf{1}_b)\rangle = \mathrm{Cov}(\varphi, Z) + \frac{d}{N}(\varphi(a) - \varphi(b))$ for every φ (Lemma 1.1(1)); apply it to $\varphi = x, y, D_{\mathrm{m}}, D_{\mathrm{a}}$ and square. □

1.5 The inertia tensor of the Magic Gem

Place unit masses at the N points $\mathbf{p}_{ij} = (x_{ij}, y_{ij}, z_{ij})$. Because all three coordinates have mean zero (Lemma 1.1(1)), the centre of mass is the origin, and the inertia tensor about it is $I = \sum_{i,j}\big(\|\mathbf{p}_{ij}\|^2 \mathrm{Id} - \mathbf{p}_{ij}\mathbf{p}_{ij}^{\mathsf{T}}\big) = N\big(\mathrm{tr}(\Sigma)\mathrm{Id} - \Sigma\big)$, where Σ is the 3×3 covariance matrix of (X, Y, Z).

Proposition 1.22 (Inertia tensor)**.** *For every arrangement* S *of* $\{1, \dots, N\}$,

$$I_{xx} = I_{yy} = \frac{n^2(n^2-1)(n^2+2)}{12}, \qquad I_{zz} = \frac{n^2(n^2-1)}{6}, \qquad I_{xy} = 0,$$
$$I_{xz} = -N\mathrm{Cov}(X,Z) = -\sum_j x_j\gamma_j, \qquad I_{yz} = -N\mathrm{Cov}(Y,Z) = -\sum_i y_i\rho_i.$$

In particular the diagonal entries are the same for all $N!$ *arrangements, and for a magic square (indeed for any arrangement satisfying the degree-*1 *conditions) the tensor is diagonal with principal moments* $\frac{n^2(n^2-1)(n^2+2)}{12}$ *(twice) and* $\frac{n^2(n^2-1)}{6}$.

Proof. $I_{xx} = \sum_{i,j}(y_{ij}^2 + z_{ij}^2) = N(\mathrm{Var}Y + \mathrm{Var}Z) = n^2\left[\frac{n^2-1}{12} + \frac{n^4-1}{12}\right] = \frac{n^2(n^2-1)(n^2+2)}{12}$ by Lemma 1.1(2), using $(n^2-1)+(n^4-1) = (n^2-1)(n^2+2)$; I_{yy} is the same by symmetry, and $I_{zz} = N(\mathrm{Var}X + \mathrm{Var}Y) = \frac{n^2(n^2-1)}{6}$. The off-diagonal entries are $-N\mathrm{Cov}(\cdot,\cdot)$ by definition; $\mathrm{Cov}(X,Y) = 0$ by Lemma 1.1(2), and $N\mathrm{Cov}(X,Z) = \sum_j x_j\gamma_j$ by Theorem 1.11(a). All three diagonal entries depend only on the multisets $\{x_{ij}\}, \{y_{ij}\}, \{z_{ij}\}$, which are the same for every arrangement. □

Remark 1.23 (What the inertia tensor does and does not detect). Proposition 1.22 is a modest statement. The principal moments of the three-dimensional Magic Gem carry no information about S beyond n, and the only arrangement-dependent entries are the two covariances $\mathrm{Cov}(X,Z)$, $\mathrm{Cov}(Y,Z)$, which are exactly the degree-1 conditions—satisfied by all 760 of the 9! arrangements for $n = 3$ and by a strictly larger-than-magic class for every n (Proposition 1.14). The deeper inertia statement is Loly's two-dimensional one (Corollary 1.13): with the *values* as masses, the planar moment of inertia is the degree-2 cross-moment $\mathrm{Cov}(Z, X^2+Y^2)$ up to an additive constant, and its invariance over all semi-magic squares is the $m = 2$ case of Theorem 1.11(a).

References

[1] Peter D. Loly. 88.30 the invariance of the moment of inertia of magic squares. *The Mathematical Gazette*, 88(511):113–119, 2004.